7. SLIDING WINDOW PROTOCOL

EX.NO: 7

Date: 30-8-26

Aim:

To write a program to implement flow control at data link layer using sliding window protocol. Simulate the flow ~~control~~ of frames from one node to another.

Sender code:

```
import time
import os
print("=" * 55)
print("    SLIDING WINDOW PROTOCOL - SENDER")
print(
window_size = int(input("Enter window size: "))
if window_size <= 0:
    print("Invalid window size !")
    exit()
message = input("Enter text message: ")
if not message:
    print("Message cannot be empty!")
    exit()
frames = []
for i, character in enumerate(message):
    frames.append({
        "frame_no": i,
        "data": character
    })
print("\nFrames created:")
for frame in frames:
    print(f"Frame No: {frame['frame_no']} DATA: {frame['data']}")
total_frames = len(frames)
base = 0
```

```
while base < total_frames:
    end = min(base + windowsize, total_frames)
    print("\n" + "_" * 55)
    print(f"sending frames {base} to {end-1}")
    with open("sender_buffer", "w") as file:
        file.write("FRAME_NO DATA\n")
        for i in range(base, end):
            file.write(
                f"{frames[i]['frame_no']} {frames[i]['data']}\n"
            )
    print("\nSender_Buffer:")
    for i in range(base, end):
        print(
            f"Frame No: {frames[i]['frame_no']}\t"
            f"DATA: {frames[i]['data']}"
        )
    print("\nwaiting for acknowledgment...")
    time.sleep(3)
    if not os.path.exists("Receiver_Buffer"):
        print("\nReceiver_Buffer not found.")
        print("Run receiver.py and process the Sender_Buffer.")
        input("Press Enter after receiver has processed the
                                                buffer...")
    try:
        with open("Receiver_Buffer", "r") as file:
            response = file.readline().strip()
    except FileNotFoundError:
        print("Receiver_Buffer not found!")
    parts = response.split()
    if len(parts) != 2:
        print("Invalid Receiver_Buffer found!")
        continue
    ack_type = parts[0]
```

```
try:
    ack_number = int(parts[i])
except ValueError:
        print("Invalid ACK/NACK number!")
        continue
print(f"\n Receiver response: {ack_type} {ack_number}")
if ack_type == "ACK":
    if ack_number == end:
            print("ACK .is as expected.")
            print("Sliding window moves forward.")
            base = ack_number
    elif base < ack_number <= end:
            print("cumulative ACK received")
            print("Moving window to frame {ack_number}.")
            base = ack_number.
    else:
            print("Unexpected ACK.")
            print("Resending current window.")
elif ack_type == "NACK":
        print(f"NACK recieved for frame {ack_number}.")
        print(f"Retransmitting frames from that frame").
        if base <= ack_number < end:
                base = ack_number
        else:
                print("Invalid NACK number.")
                print("Resending current window")
    else:
        print("Invalid response received")
    time.sleep(1)
print("\n
print("All frames transmitted successfully!")
print("Original message: {message}")
```

Output :

```
SLIDING WINDOW PROTOCOL - SENDER
Enter window size : 3
Enter test message : HELLO
```

Sender Buffer:

Frame No: 0 DATA: H

Frame No: 1 DATA: E

Frame No: 2 DATA: L

Waiting for acknowledgment...

Receiver response: ACK 3

ACK is as expected

Sliding window moves foward

Sender-Buffer:

Frame No: 3 DATA: L

Frame No: 4 DATA: O

Waiting for acknowledgment...

Receiver response: ACK 5

ACK is as expected.

Sliding window moves forward.

All frames transmitted successfully!

Original message: HELLO.

Receiver code:

```
import os
SENDER_FILE = "sender_buffer.txt"
RECEIVER_FILE = "receiver_buffer.txt"
expected_frame = 0
while True:
    if not os.path.exists(SENDER_FIE):
        print("Sender_Beffer not found!")
        break
    with open(SENDER_FILE, "r") as file:
        lines = file.readlines()
    if len(lines) == 0:
        print("No frames available")
        break
print("In Frames received")
error = False
error_frame = None
```

```
for line in lines:
    parts = line.strip().split(",")
    frame_no = int(parts[0].split(":")[1].strip())
    data = parts[1].split(":")[1].strip()
    print(f"Frame No: {frame_no}, DATA: {data}")
    if frame_no != expected_frame:
        error. = True.
        error_frame = frame_no
        break.
    expeted_frame += 1
if error:
    with open(RECEIVER_FILE, "w") as file:
        file.write(f"NACK: {error_frame}")
    print(f"\nIncorrect frame No. {error_frame}")
    print(f"NACK: {error_frame}")
else:
    with open(RECEIVER_FILE, "w") as file:
        file.write(f"ACK: {expected_frame}")
    print(f"\nAll frames received correctly")
    print(f"ACK: {expected_frame}")
```

Output:

```
Frames Received
frame no: 0     Data: H
frame no: 1     Data: E
frame no: 2     Data: L
All frames received correctly.
ACK: 3.
```

Result:

Thus, the sliding window protocol was implemented successfully using python, and the flow of frames b/w the sender and receiver was simulated using ACK/NACK.